

Basic of Internet

Lec 8

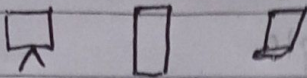
OSI Model - in depth

Date 26/9/24
Page

→ MAC Address / Physical Address

Medium Access Control

- doesn't change (can change but illegal)



every device has
mac address

- How many mac address?

- no. of nic - that much mac address
bno. of ways device can
connect with internet.

- MAC Address Format

↓
48 bits
(hexadecimal)

Some id in
(then company
in other
company
another id)

Physical Address
MAC
Rohan
Rohu
Chotu

00:00:00:00:00:00
ff:ff:ff:ff:ff:ff

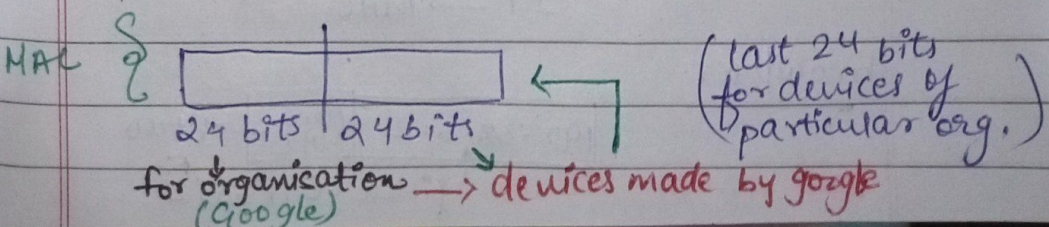
8bit for one

15 6

IP Add.

$$6 \times 8 = 48 \text{ bits}$$

How MAC address is distributed?



$2^{24} \rightarrow$ organisations

$2^{24} \rightarrow$ for devices of their org.
(manufacturing)

24
bits

{ CC:46:D6 - Cisco
3C:5A:B4 - Google

- Three ways of formats

hyphen-hexadecimal

00-0a-83-b1-c0-8e

colon-hexadecimal

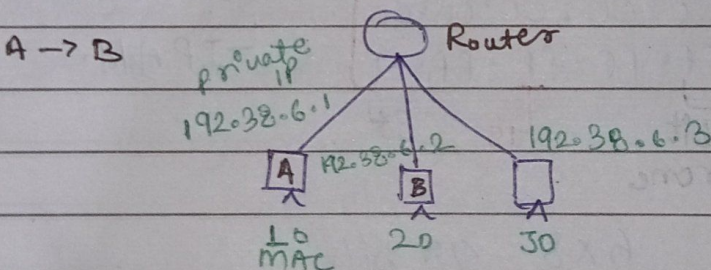
00:0a:83:b1:c0:8e

period separated hexadecimal

000.a83.b1c.08e

- Data Link layer (further continued)

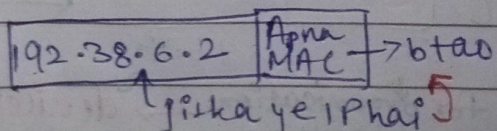
in LAN how other devices MAC
address is known?



ARP: Address Resolution Protocol

IP → MAC

broadcast



20

ans

→ 192.38.6.1

(sender) of broadcast

Then,

10 | Data | 20

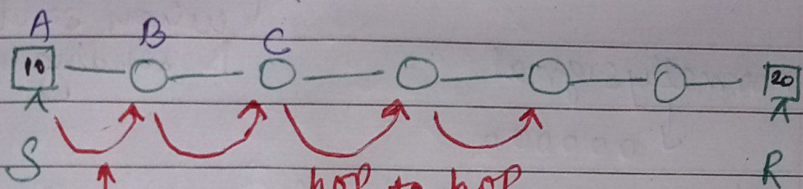
one time known - IP address help.

broadcast

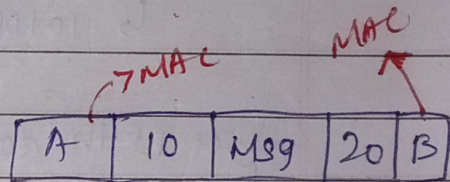
23.1.2.2 | Data | 23.255.255.255

Sender address

broadcast address

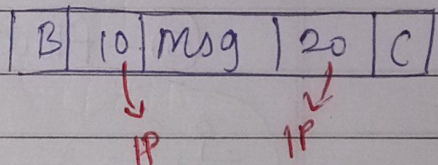


MAC Address needed of router



letter

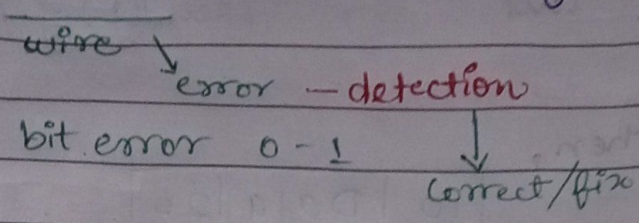
Sender add. delhi
receiver add. mumbai



[D] → [hr] → [up] → [MP] → [Mu]

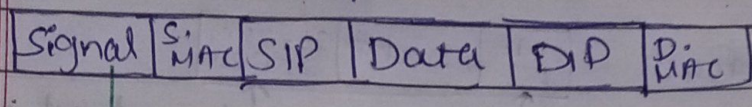
When some illegal activity happens, ISP also stores mac-address bcoz IP address can be changed when disconnected from internet.

③ - error detection & handling



④ - framing

Structure in which data travels



how receiver knows it have to read data now?

normally signal
↓
000000

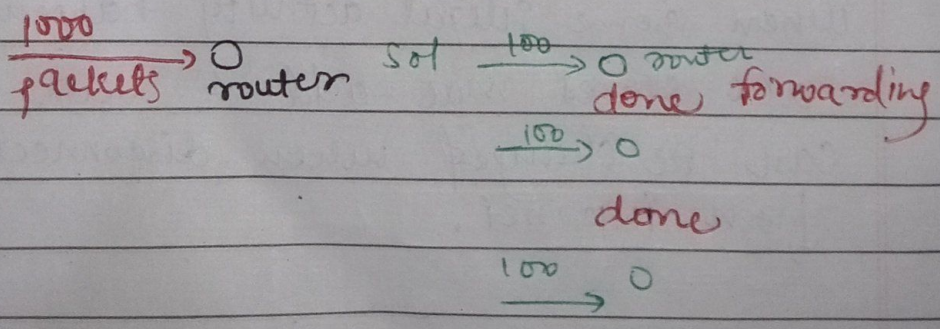
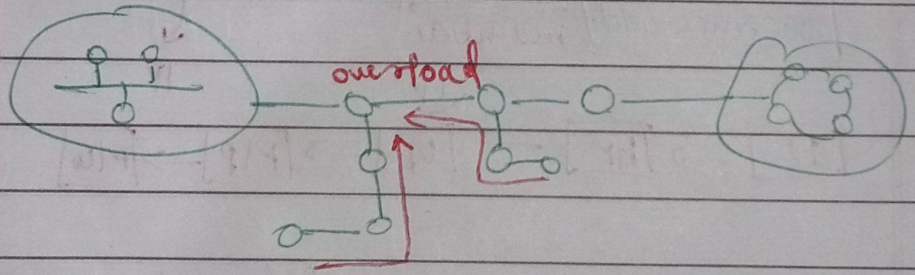
when sender is sending msg.

when msg sent signal

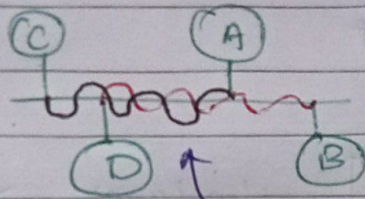
↳ 101010101010 --- 11 [msg] Tsend

after 11 the data is read.

⑤ - flow control



(6) - access control



Signal
& data collide

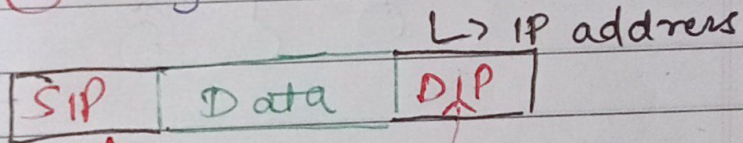
! data can corrupt !

Sol: one data transfer at a time.

DLL → whoever to provide access

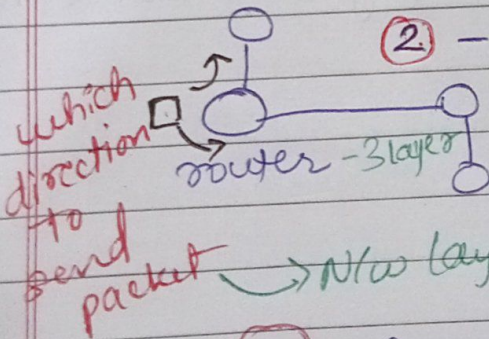
3. Network layer

(1) - Logical Address



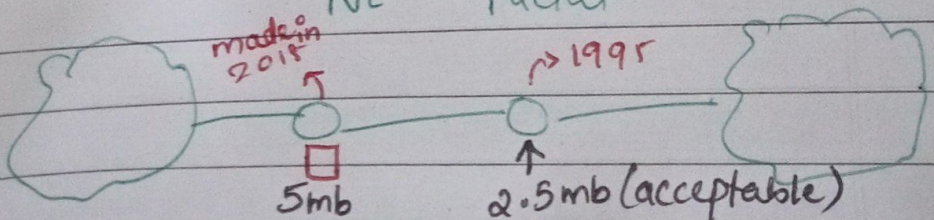
N/w Layer

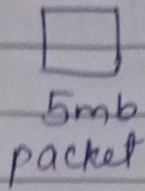
(2) - routing



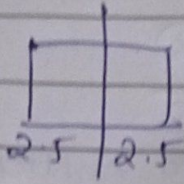
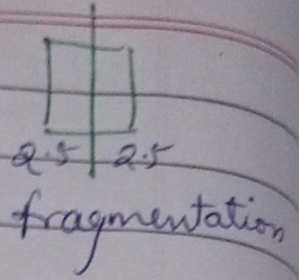
(3) - fragmentation & reassembly

DLL - frame
NL - Packet) same

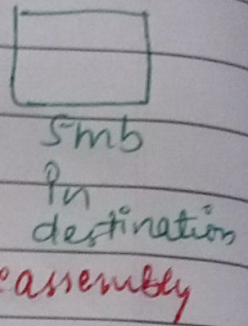




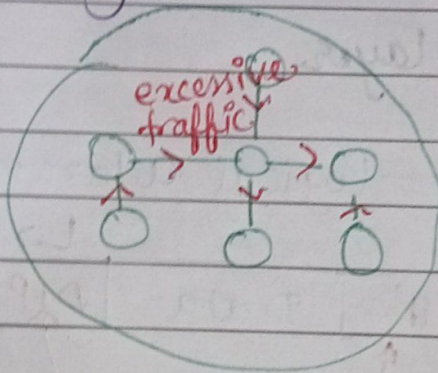
N/w layer →



N/w layer →



④ - congestion control



- speed slow
- n/w layer handler that
- limit users/packet

bandwidth → speed of data travel (10mbps)